FSCNet: Frequency-Spatial Collaborative Learning for Segmentation of Oceanic Internal Solitary Waves From Geostationary Satellite Imagery

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***Abstract*—Extracting internal solitary waves (ISWs) from optical remote sensing imagery is constrained by strong background interference, severe foreground sparsity, and insufficient utilization of frequency-domain features. A frequency-spatial collaborative network (FSCNet) is proposed for pixel-level segmentation using Fengyun-4 (FY-4) geostationary satellite imagery. FSCNet introduces an adaptive spectrum attention module (ASAM), which decomposes the input image into directional, high-frequency, and low-frequency components in the Fourier domain and adaptively recalibrates their contributions under varying ocean background conditions. This design enhances anisotropic wave textures while suppressing isotropic random noise and background clutter. A multilevel feature synergy architecture couples a lightweight Transformer for global dependency modeling with a bidirectional feature pyramid network (BiFPN) for efficient cross-scale aggregation, enabling robust capture of large-scale wave propagation trends and fine-grained local textures. To sharpen edge predictions and preserve wave-skeleton integrity, a boundary-aware module (BAM) and multitask loss are incorporated to reduce boundary ambiguity. A standardized FY-4 ISW dataset with rigorous pixel-level annotations is constructed. Experimental results demonstrate that FSCNet achieves a Dice score of 0.6418 and a mean intersection over union (mIoU) of 0.7352, outperforming representative segmentation methods. Additional zero-shot validation on Himawari-9 imagery further indicates the cross-region and cross-sensor generalization potential of FSCNet.**

***Index Terms*—Frequency-spatial learning, geostationary satellite imagery, internal solitary waves (ISWs), semantic segmentation.**

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I. Introduction

INTERNAL solitary waves (ISWs), as a typical submesoscale nonlinear wave phenomenon [1], are widely distributed across global continental shelves and marginal seas [2]. Characterized by large amplitudes and strong nonlinearity [3], these waves can propagate over long distances across deep ocean basins [4], playing a crucial role in the vertical mixing of upper ocean waters, energy cascade processes, and material transport [5]. The generation and evolution of ISWs not only profoundly influence the dynamic environment of marine ecosystems but also pose potential threats to marine engineering and military activities, such as offshore drilling platforms, underwater vehicle navigation, and underwater acoustic communication [6]. However, constrained by the transient nature and nonstationarity of ISWs across spatiotemporal scales, traditional in situ observation methods struggle to capture their large-scale dynamic evolution processes. Therefore, developing high-precision and automated techniques for interpreting ISW signatures based on remote sensing data holds significant scientific importance and application value. These advancements are essential for enhancing marine environmental situational awareness, improving disaster warning capabilities, and deepening the understanding of ISW dynamic mechanisms.

With the rapid advancement of Earth observation technologies, satellite remote sensing has emerged as the primary tool for monitoring global ISW activities. Synthetic aperture radar (SAR), leveraging its day-and-night and all-weather imaging capabilities, captures alternating light and dark stripes modulated by ISW flow fields by detecting microscale roughness variations on the sea surface [7], thus serving as a vital data source for ISW observation. However, the backscattering signatures of ISWs in SAR imagery are prone to confusion with complex sea-surface features such as ship wakes, oceanic fronts, and atmospheric gravity waves [8], and oceanic vortex streets [9]. Furthermore, these signatures are significantly influenced by imaging geometry and sea-state conditions, thereby complicating the task of automated interpretation.

In contrast, optical remote sensing directly reflects sea-surface roughness anomalies induced by ISWs by capturing variations in reflectance within the sunglint region [2], [10]. Compared with polar-orbiting satellites and SAR, which are

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characterized by relatively long revisit cycles, Fengyun-4 (FY-4) geostationary satellite optical imagery provides a unique advantage in high-frequency and fixed-region continuous observation. With a temporal resolution of up to 15 min, FY-4 enables continuous monitoring of the generation, propagation, and temporal evolution of oceanic ISWs, which are highly dynamic and may undergo rapid morphological changes within a short period. This high-frequency sampling capability allows geostationary satellites to capture dynamic features that are often missed by traditional polar-orbiting platforms.

Notably, with the high-frequency observations provided by geostationary satellites, visible-light imagery offers dynamic views of ISW propagation [11]. Under suitable illumination and sunglint conditions, it can clearly reveal alternating bright and dark stripe patterns associated with ISWs, providing physically interpretable texture and boundary cues for pixel-level segmentation. Although optical imaging is constrained by cloud cover and illumination conditions, it encapsulates rich textural and spectral information. Under clear-sky conditions, it delineates the geometric structure of ISW crest lines with superior clarity, offering an irreplaceable observational perspective for investigating the fine morphological characteristics of ISWs.

In recent years, deep learning has witnessed significant progress in the fields of remote sensing image segmentation, object detection, and parameter inversion [12], and unsupervised image superresolution [13], [14]. Classical convolutional neural networks (CNNs) and their derivative architectures, such as U-Net [15], SegNet [16], and the DeepLab series [17], have been widely applied to achieve pixel-level segmentation of oceanic ISWs from SAR and optical remote sensing imagery, due to their powerful capabilities in feature extraction and spatial context modeling. These methods automatically learn texture, edge, and shape features of ISWs in the spatial domain, demonstrating significant improvements in both efficiency and accuracy compared to traditional methods that rely on handcrafted features. However, oceanic ISWs are intrinsically physical fluctuations characterized by specific wavenumbers and directional spectra. Their most prominent spatial feature, the periodic stripe pattern, manifests as energy-concentrated signals in the frequency domain. Existing methods based on spatial domain convolutions, while effective in capturing local textures, still suffer from limitations in explicitly modeling this global periodic structure, separating wave features of different scales, and suppressing sea-surface interference within similar frequency bands. Therefore, incorporating frequency-domain information into deep learning frameworks, specifically by analyzing low-frequency components (corresponding to large-scale waveforms) and high-frequency components (corresponding to fine edges) in the spectrum, holds promise for characterizing the physical properties of ISWs more fundamentally. This approach facilitates more robust and accurate signature extraction against complex sea-surface backgrounds.

Against this research background, this article addresses the challenges associated with the precise extraction of ISW signatures from optical remote sensing imagery, the insufficient utilization of frequency-domain features, and the limited adaptability of existing methods to complex sea-surface backgrounds. To this end, we propose a deep learning segmentation model based on frequency-domain information enhancement. Utilizing visible-light remote sensing imagery acquired by China's FY-4 series geostationary satellites as the primary data source, our method explicitly incorporates the periodicity and directionality features of ISWs in the frequency domain. By effectively fusing spatial and frequency-domain information, the proposed model achieves high-precision pixel-level segmentation of ISW signatures. The main contributions of this article can be summarized as follows.

1) *Multidomain Feature Representation:* A mechanism based on multidomain collaborative enhancement is proposed. By constructing a high-order frequency-aware matrix, it explicitly decouples low-frequency trends and high-frequency details, effectively embedding physical priors to enhance discrimination against complex backgrounds.
2) *Multilevel Feature Synergy Architecture:* An efficient architecture integrating global dependency modeling with a bidirectional feature pyramid network (BiFPN) for deep semantic fusion is designed. A boundary-aware constraint mechanism and a multitask loss are incorporated to enforce skeletal integrity and precise localization under noisy conditions.
3) *Development of a Specialized Dataset:* A specialized ISW dataset is constructed using multitemporal FY-4A/B imagery. Featuring rigorous pixel-level annotations across diverse sea states, it provides a robust benchmark for validating optical ISW extraction algorithms.

The remainder of this article is organized as follows. Section II reviews the state of the art in related fields. Section III details the overall architecture and key module designs of the proposed model. Section IV presents the experimental results and comparative analysis, followed by an in-depth discussion of model performance. Finally, Section V concludes this article and outlines future research directions.

II. Related Works

A. Deep Learning for ISW Segmentation

In satellite remote sensing imagery, oceanic ISWs typically manifest as arc-shaped or linear stripe packets characterized by high nonlinearity [18], [19]. Traditional automatic extraction methods predominantly employ edge detection operators or wavelet transforms [20], [21], [22]. Although these methods are capable of capturing local intensity gradient variations, they often struggle to differentiate ISW stripes from complex sea-surface background noise, such as ship wakes and atmospheric gravity waves [8], resulting in insufficient robustness in low signal-to-noise ratio (SNR) environments [22], where enhanced deep image priors are often required to reconstruct signals from degraded observations [23].

With the evolution of deep CNNs, the encoder–decoder-based semantic segmentation paradigm has become the

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dominant technique for ISW extraction. To mitigate the interference of ocean background noise on weak signals, Zhang et al. [24] incorporated a channel attention mechanism (SE Block) into IWE-Net, effectively improving the representation capability of ISW features in multisource heterogeneous imagery. Similarly, Barintag et al. [25] enhanced the model's ability to capture texture details in SAR imagery through a nested U-structure (MTU^2-Net). However, despite the remarkable performance of these improved models in local feature extraction, they are constrained by the inherent local receptive field of convolution operators. Consequently, when processing ISW stripes characterized by large-scale continuous topological structures, phenomena such as signature fragmentation and discontinuity frequently occur.

To tackle the issue of missing long-range dependencies, researchers have begun to introduce Transformer architectures. Xu et al. [26] leveraged the self-attention mechanism of TransUNet to achieve continuous segmentation of ISWs in the East China sea over long time series, while Zhang et al. [27] validated the advantages of SegFormer in preserving the geometric structures of ISWs without the constraints of positional encoding. However, existing deep learning paradigms-based purely on the spatial domain still face challenges when processing geostationary visible-light imagery. On the one hand, subject to nonlinear interference from sunglint and cloud shadows, pure spatial convolution operations struggle to effectively separate weak ISW textures in low SNR environments. On the other hand, these methods overlook the inherent frequency-domain fingerprints of ISWs. This results in a lack of discriminative power in suppressing morphologically similar background clutter. Furthermore, the downsampling mechanisms in deep networks inevitably lead to the loss of high-frequency edge details.

B. Frequency-Domain Learning in Visual Tasks

Addressing the inherent limitations of spatial convolution in capturing global texture patterns and long-range dependencies, frequency-domain-based feature learning has emerged as a significant breakthrough for the refined processing of remote sensing imagery in recent years [28], [29]. Viewed from a hydrodynamic perspective, oceanic ISWs intrinsically manifest as quasiperiodic stripe structures. These highly ordered coherent structures exhibit remarkable energy compactness in the spectrum of the Fourier Transform. As shown in Fig. 1(b) and (c), although simple high-pass filtering enhances edges, it inevitably introduces high-frequency speckle noise, whereas low-pass filtering results in oversmoothing and the loss of fine texture details. In contrast, Fig. 1(d) demonstrates that directional bandpass filtering effectively isolates the skeletal structures of ISWs, mapping them to distinct high-energy regions in the spectrum while suppressing background clutter. This reveals that the discriminative fingerprints of ISWs are not merely distributed across high or low frequencies but are concentrated within specific directional energy bands.

Early attempts primarily focused on introducing fixed frequency-domain transformations at the input stage of the network. For instance, the FDCNN proposed by Song et al. [30] utilizes spectral saliency maps to maintain global consistency

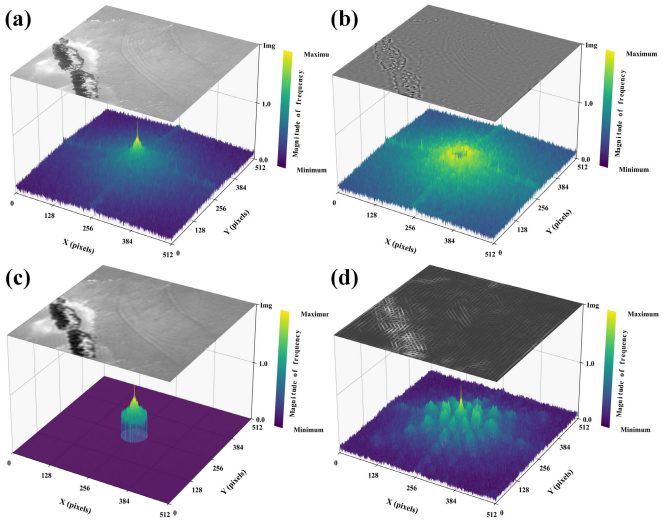


Fig. 1. Visualization of the ISWs in spatial and frequency domains. (a) Original image and its spectrum. (b)–(c) Results of high-pass and low-pass filtering, illustrating the tradeoff between noise introduction and feature blurring. (d) Directional filtering result, demonstrating that ISW signatures are concentrated in specific directional frequency bands.

in classification. Although such methods leverage global information, their reliance on static filters makes it difficult to adapt to the complex and variable marine environment. With the evolution of feature extraction techniques, the field of computer vision has begun to explore the deep integration of frequency-domain transformations into network architectures. From the perspective of compressed sensing, Qin et al. [31] proposed FcaNet, demonstrating that global average pooling (GAP) is merely a special case of the discrete cosine transform. By employing multispectral energy recalibration, they significantly enhanced the network's texture perception capabilities. Inspired by this, Rao et al. [32] further introduced GFNet, which utilizes the fast Fourier transform (FFT) to perform global token mixing in the frequency domain, achieving efficient long-range dependency modeling with loglinear complexity.

In the field of remote sensing, recent efforts have been dedicated to bridging the semantic gap between spatial and frequency-domain features by constructing dual-domain collaborative architectures. Zhang et al. [33] designed a dual-domain spatial-frequency detector with boundary awareness, which incorporates a dual-domain mixer to acquire a global receptive field via the FFT, thus complementing the CNN spatial branch. Similarly, the GFDNet proposed by Zhou et al. [34] dynamically captures contextual information by designing a learnable global filter. Gao et al. [35] recently conducted advanced exploration in the field of general remote sensing semantic segmentation, proposing the adaptive frequency enhancement network. The core of this model lies in the design of the adaptive frequency and spatial feature interaction module, which is capable of dynamically generating high- and low-frequency spectrum masks based on the semantic content of the input imagery. By adaptively separating and modulating feature components across different frequency bands, this mechanism effectively enhances the

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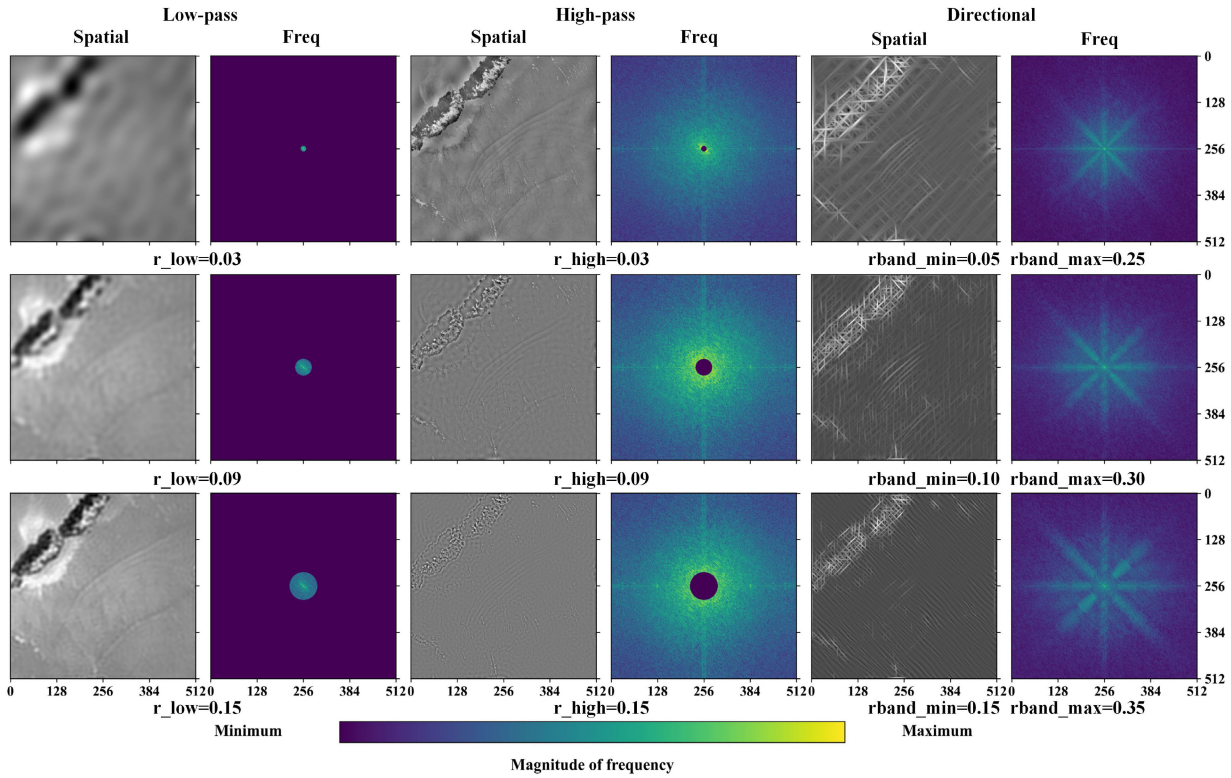


Fig. 2. Sensitivity analysis of ISW feature extraction using fixed frequency-domain filters. The visualization displays spatial reconstructions and corresponding spectra under varying cut-off frequencies (r_{low}, r_{high}) and directional bandwidths (r_{band}).

interaction capabilities between spatial and frequency-domain features. To a certain extent, it addresses the limitation of traditional fixed filters in adapting to the scale variations of ground objects.

However, although the aforementioned data-driven adaptive mechanisms perform excellently in land remote sensing scenarios, they encounter severe challenges when processing high-dynamic oceanic ISW observations. As visually quantified in Fig. 2, the separation of ISW features faces a precarious tradeoff due to severe spectral aliasing with background clutter. A marginal deviation in the cutoff threshold leads to distinct failure modes: a lower threshold fails to suppress sea-surface noise, while a slightly higher threshold causes the complete structural erosion of weak wave textures.

Existing adaptive methods predominantly rely on implicit data-driven learning, lacking explicit modeling of the physical fingerprints of oceanic ISWs. For instance, ISWs possess distinct directional energy concentration and specific scale distributions in the frequency domain. Due to the failure to explicitly decouple and leverage these physical priors, existing models often struggle to strike an optimal balance between suppressing isotropic noise and preserving anisotropic signals under complex sea conditions.

C. Multiscale Fusion and Boundary Refinement

ISWs exhibit extremely complex geometric morphologies in remote sensing imagery, with their spatial spans extending from hundreds of meters to tens of kilometers. This pronounced multiscale characteristic demands that segmentation models possess robust capabilities for spatial context aggregation [36]. Traditional single-scale CNNs often suffer from an irreversible loss of low-level, high-resolution geometric details when extracting deep semantic information via downsampling. Consequently, this leads to blurred ISW edges or the loss of fine stripes, failing to meet the requirements for refined observation.

To alleviate this issue, multiscale feature fusion mechanisms have become pivotal for enhancing the performance of object detection in remote sensing. The feature pyramid network (FPN) [37], by constructing a top-down feature pyramid, achieved the effective integration of high-level semantic information with low-level positional information for the first time. Building upon this, to address the drastic variations in object scale within remote sensing imagery, researchers have further proposed the path aggregation network (PANet) and the BiFPN [38]. By introducing weighted bidirectional cross-scale connections, BiFPN achieves efficient reuse and deep fusion of multilevel features. It has been demonstrated to possess significant advantages in the extraction of small-scale targets and weak features amidst complex sea-surface backgrounds [39].

However, for ISWs, linear elongated targets characterized by extremely high aspect ratios, relying solely on feature fusion is often insufficient to maintain the topological continuity of their signatures. This is particularly evident in the presence of strong noise interference, where signature fragmentation frequently occurs. Consequently, boundary-aware learning has gradually emerged as a research hotspot for refined

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segmentation. By integrating dedicated boundary detection branches or attention modules, such as CBAM [40], into the backbone network and combining them with multitask learning strategies, models can explicitly learn the contour features of targets. This reinforces the constraints on edge pixels amidst complex background noise. Although such methods have achieved significant progress in tasks like medical imaging, such as skin lesion segmentation [41], and road extraction, they remain inadequate in the field of oceanic ISWs. Because ISW crest-line edges are often accompanied by diffuse intensity transitions and suffer from severe interference by sea-surface clutter, including ship wakes and atmospheric gravity waves [8], existing boundary optimization mechanisms struggle to precisely localize the skeletal structure of ISWs. Therefore, combining cross-scale feature evolution with robust boundary-aware mechanisms tailored to the marine environment is an essential path toward achieving high-consistency extraction of ISW signatures.

III. Materials and Methods

This section systematically details the dataset construction and the proposed frequency-spatial collaborative network (FSCNet). First, the data source from FY-4 series geostationary satellites and the rigorous data production process are introduced. Subsequently, tailored to the specific imaging characteristics of these data, an end-to-end segmentation architecture is established to synergize adaptive frequency domain filtering with spatial global modeling.

A. Dataset Construction and Preprocessing

1) Region of the Interest Description: To construct an ISW dataset featuring physical representativeness and morphological diversity, the selection of experimental regions focuses on maritime areas characterized by frequent ISW activities and clear optical textures within geostationary satellite imagery. Guided by this principle, this study identifies the Sulu sea and the Sulawesi sea, situated in the marginal seas of the Western Pacific, as the core observation zones.

As globally recognized hotspots for ISW research [2], these two maritime regions are situated at the confluence of tropical currents. Driven by the combined forces of strong tidal energy and complex submarine topography, they exhibit an extremely high frequency of ISW generation and significant spatial extensibility. Specifically, ISWs in the Sulu sea are typically excited by strong tidal currents traversing shallow topography, such as the Pearl Bank or the Sibutu Passage; their amplitudes can reach tens of meters, with propagation distances frequently exceeding one hundred kilometers [42], [43]. In contrast, the Sulawesi sea, modulated by tidal currents from complex archipelago channels and the Makassar Strait, exhibits distinct seasonal variability in its ISW activity [44], often forming clear strip-like structures along deep-sea slopes or island margins [45].

Owing to the abundant tidal energy and stable water stratification in this region, the aforementioned physical processes manifest as highly distinctive visual signatures in remote sensing imagery. Particularly in cloud-free visible-light imagery acquired by FY-4 series satellites, ISW stripes typically appear as elongated, high-contrast reflectance bands with alternating bright and dark patterns. These clear optical textures not only facilitate the capture of the fine structural details of ISWs but also provide an ideal natural testbed for constructing high-quality semantic segmentation datasets.

The Sulu sea and the Sulawesi sea are not intended to fully represent all global marine environments. Instead, they are selected as physically typical and observationally challenging regions for ISW segmentation, as they contain frequent ISW events, diverse wave morphologies, complex island topography, and variable sea-surface backgrounds. Therefore, the representativeness of this dataset mainly lies in its coverage of typical ISW generation mechanisms and remote-sensing imaging characteristics in tropical marginal seas. To further examine whether the trained model can generalize beyond these two regions, an additional cross-region and cross-sensor validation experiment is conducted in Section IV-E using Himawari-9/Advanced Himawari imager (AHI) imagery over the Lombok Strait.

2) Data Construction: Visible-light remote sensing imagery, as a vital instrument for observing oceanic ISWs, has long occupied a significant position in academic research. However, compared to active remote sensing modalities such as SAR, high-quality visible-light remote sensing datasets remain relatively scarce. This scarcity is primarily attributed to the stringent dependence of optical imaging on meteorological and illumination conditions: it is only during a transient window characterized by clear skies, low cloud cover, and appropriate solar angles that the faint optical textures of the sea surface can be clearly recorded. These dual spatiotemporal constraints have resulted in existing public samples being extremely sparse and lacking standardization, thereby significantly impeding the development of data-driven algorithms for oceanic ISW signature segmentation.

In light of this, this study constructs a visible-light remote sensing dataset based on observational data from China's geostationary meteorological satellites, FY-4A and FY-4B. The advanced geostationary radiation imager (AGRI) sensor onboard this satellite series possesses high-temporal resolution and multispectral observation capabilities, enabling the continuous acquisition of full-disk Earth observation data at a spatial resolution of 500 m. Specifically, Band 2 of AGRI was selected in this study. This band is a visible red band with a central wavelength of 0.65 μm and a spatial resolution of 500 m, which is suitable for observing sea-surface reflectance variations associated with internal ISWs.

The construction process of the dataset follows strict standardized protocols, including raw data preprocessing, spatial cropping, sample filtering, pixel-level annotation, and annotation quality control. First, an automated preprocessing pipeline was established to perform decoding and visualization mapping on the FY-4 full-disk raw data from June 2022 to September 2023. Based on geographic coordinates, the imagery was then precisely cropped to retain only the regions of interest, including the Sulu sea and the Sulawesi sea. Images severely contaminated by clouds, land occlusion, or unclear sea-surface textures were removed through visual

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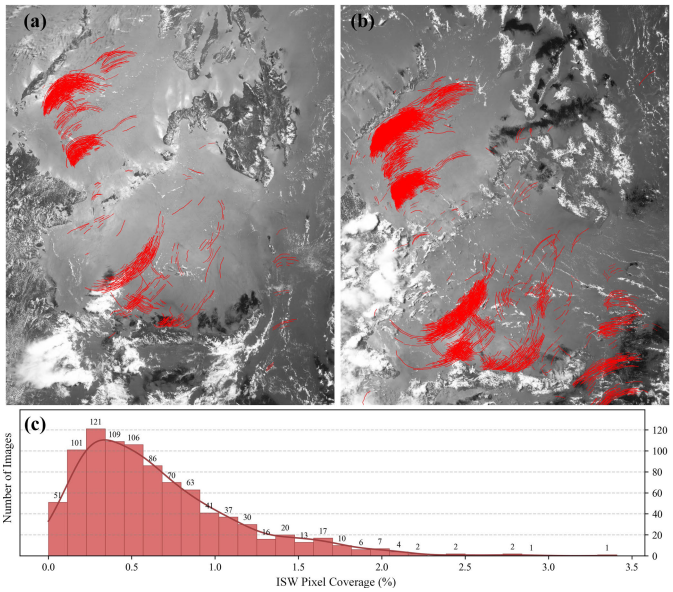


Fig. 3. Visualization of the constructed FY-4 ISW dataset and its spatial distribution. (a) and (b) Annotated samples acquired from the FY-4A and FY-4B satellites, respectively, both covering the Sulu sea and Sulawesi sea observation regions. (c) Histogram of ISW pixel coverage percentages with frequency values annotated atop each bar. The red smooth curve represents the kernel density estimation, explicitly highlighting the long-tail morphology of the distribution.

inspection and quality filtering. This process ultimately yielded 916 high-quality images, including 248 samples from FY-4A and 668 samples from FY-4B.

To generate reliable ground truth masks, we adopted a two-stage multiannotator cross-validation protocol. Before formal annotation, two researchers with experience in marine remote sensing interpretation and ISW identification jointly established a unified physical interpretation standard. In optical imagery, ISWs usually appear as alternating bright and dark stripes within sunglint regions; therefore, annotation boundaries were mainly determined according to local gradient changes, stripe continuity, and wave-train morphology. The two annotators then independently annotated the same set of images at the pixel level, and the intersection over union (IoU) between the corresponding masks was calculated to evaluate interannotator consistency.

When the IoU between two masks was not lower than 95%, the final ground truth was generated using a conservative pixelwise intersection strategy, retaining only the pixels identified as ISWs by both annotators. When the IoU was lower than 95%, the two annotators jointly re-examined the original image and refined the disputed regions until a consensus annotation was obtained. The 95% threshold was determined based on a pilot statistical analysis of 50 randomly selected images, which showed that ISW pixels are highly sparse in 512×512 images, with 74% of the samples containing fewer than 2000 ISW pixels. This annotation protocol combines quantitative consistency evaluation with physically based visual interpretation, thereby improving the objectivity and reliability of the generated ground truth masks.

For the model development and evaluation phases, the dataset is systematically partitioned into training and testing sets using an 8:2 split ratio, yielding 733 training samples and 183 independent testing samples. Furthermore, statistical analysis of these annotations reveals a severe class imbalance. The ISW foreground pixel coverage per image ranges from 0.00% to 3.41%, with an average coverage of only 0.65%. This pronounced imbalance poses a significant challenge for pixel-level classification, directly motivating our incorporation of boundary-aware loss functions to prioritize hard-to-segment wave skeletons. Representative samples and coverage distribution statistics are visualized in Fig. 3, showcasing the diversity of ISW signatures and the extreme sparsity of the targets.

B. Overall Architecture of FSCNet

The overall architecture of FSCNet is illustrated in Fig. 4. The model first feeds geostationary satellite imagery into the adaptive spectrum attention module (ASAM) to extract specific frequency-domain features. Subsequently, these features are encoded by the EfficientNet backbone and a lightweight Transformer to simultaneously capture local patterns and global spatial dependencies. The encoder outputs hierarchical feature maps C_1–C_5, which are further fed into the BiFPN to perform bidirectional multiscale feature fusion and generate the corresponding feature representations M_1–M_5. Among them, the highest resolution fused feature M_1 retains fine spatial gradients and boundary details, while M_2–M_5 provide richer semantic information with larger receptive fields. Instead of directly concatenating all feature maps in the primary segmentation path, FSCNet adopts a boundary-guided decoupled design. Specifically, M_2–M_5 are used in the primary segmentation branch for multiscale semantic aggregation, whereas M_1 is routed into the boundary-aware module (BAM) to extract refined boundary features. Finally, the semantic features from M_2–M_5 and the boundary-aware features from BAM are integrated before the final segmentation head, thereby generating high-precision segmentation masks.

1) Multidomain Feature Injection and Encoding Stage: To overcome the limitation of limited information content in single-channel visible-light imagery, the model first feeds the original grayscale image into the ASAM. Through FFT and adaptive filtering, three auxiliary channels containing high-frequency, low-frequency, and directional energy information are generated. These auxiliary channels are subsequently concatenated with the original image to form a four-channel tensor, which is then input into the backbone network. We employ EfficientNet as the encoder. Benefiting from its compound scaling strategy, EfficientNet systematically balances depth, width, and resolution, thereby reconciling computational efficiency with representation capability. Furthermore, a lightweight Transformer module is introduced between the top layer C_5 of EfficientNet and the lateral convolution layers. This design aims to capture long-range spatial dependencies beyond local convolutional operations, establishing global contextual perception of ISW stripes. After encoding, the backbone produces multiscale features with complementary characteristics: shallow features retain high-resolution spatial details and local gradients, while deeper features provide stronger semantic representations and broader contextual information.

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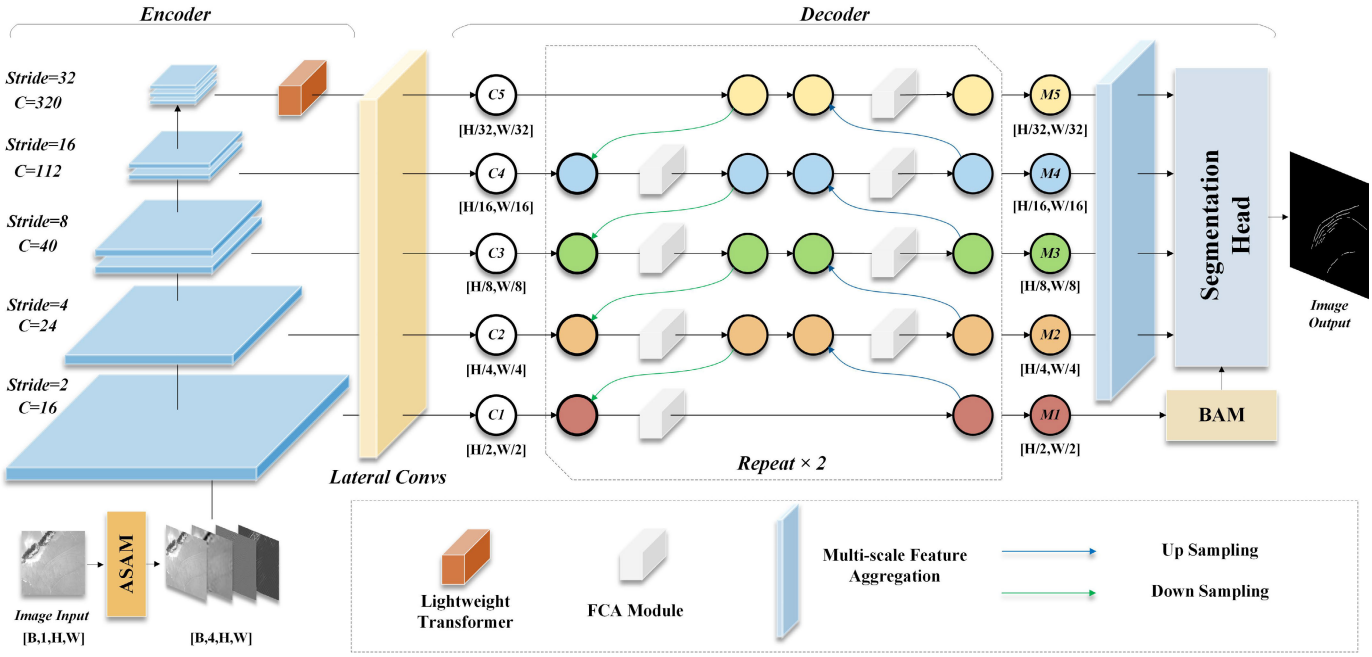


Fig. 4. Overview of the proposed FSCNet. The architecture follows an encoder–decoder paradigm integrated with three core components: the ASAM for explicit frequency-domain decoupling, the BiFPN for cross-scale feature aggregation, and the BAM for refined skeleton reconstruction.

2) Multiscale Feature Collaborative Fusion Stage: The decoder employs BiFPN to bridge the gap between high-level semantics and low-level spatial details during feature fusion. In contrast to traditional unidirectional pathways, BiFPN achieves efficient weighted fusion of the multilevel features C_1–C_5 output by the encoder through learnable cross-scale connections and iterative bidirectional information propagation, including both top-down and bottom-up pathways. This mechanism enables the model to integrate deep semantic information with shallow high-resolution geometric textures, thereby improving the representation of both global context and fine spatial details.

After bidirectional fusion, BiFPN generates multiscale fused representations M_1–M_5. The features M_2–M_5 are used for multiscale semantic aggregation in the primary segmentation branch, whereas M_1, which retains the highest spatial resolution, is routed to the BAM to provide fine boundary localization cues for the final prediction.

3) Boundary-Aware and Multitask Output Stage: For the precise delineation of fine ISW skeletal structures, FSCNet adopts a dual-path output design consisting of a primary segmentation path and an auxiliary boundary path. In the primary path, the resolutions of the semantic features M_2–M_5 are first aligned and then concatenated to obtain a semantically enriched representation. In the auxiliary path, the highest resolution feature M_1 is processed by BAM to extract boundary-aware features and generate a boundary prediction map.

The boundary-aware features derived from M_1 are further fused with the semantic representation aggregated from M_2–M_5 before being fed into the final segmentation head. Through the joint optimization of segmentation and boundary prediction, FSCNet enhances edge sharpness and topological continuity while reducing potential spatial offsets between predicted boundaries and actual ISW locations.

C. Multidomain Representation and Spectral Augmentation

Considering the characteristics of ISWs in visible-light imagery, specifically their weak signals and susceptibility to interference from complex sea conditions, we construct a multidomain feature enhancement frontend. This module is designed to fundamentally enhance the model's perception capability regarding the physical features of ISWs. It primarily consists of the following two key components.

1) Heterogeneous Frequency Channel Construction and Adaptive Enhancement: Based on the hydrodynamic characteristics of ISWs, the periodic sea-surface roughness variations induced by ISWs exhibit specific energy distributions in the frequency domain. To explicitly exploit this property while preserving the original spatial information, the ASAM adopts a parallel feature generation architecture, as illustrated in Fig. 5. One branch directly preserves the original single-channel grayscale image, while the other branch extracts and adaptively modulates frequency-domain components.

Given an input grayscale image $I \in \mathbb{R}^{1\times H\times W}$, ASAM first transforms it into the frequency domain using the FFT. To facilitate the construction of predefined frequency-domain filters, the zero-frequency component is shifted to the center of the spectrum

$$S(u,v) = \text{Shift}(\mathcal{F}(I)) \tag{1}$$

where $\mathcal{F}(\cdot)$ denotes the FFT operation, Shift$(\cdot)$ denotes the frequency-centering operation, and $S(u,v)$ represents the shifted frequency spectrum of the input image. Then, three predefined frequency-domain filters, including a directional bandpass filter H_d, a high-pass filter H_h, and a low-pass filter

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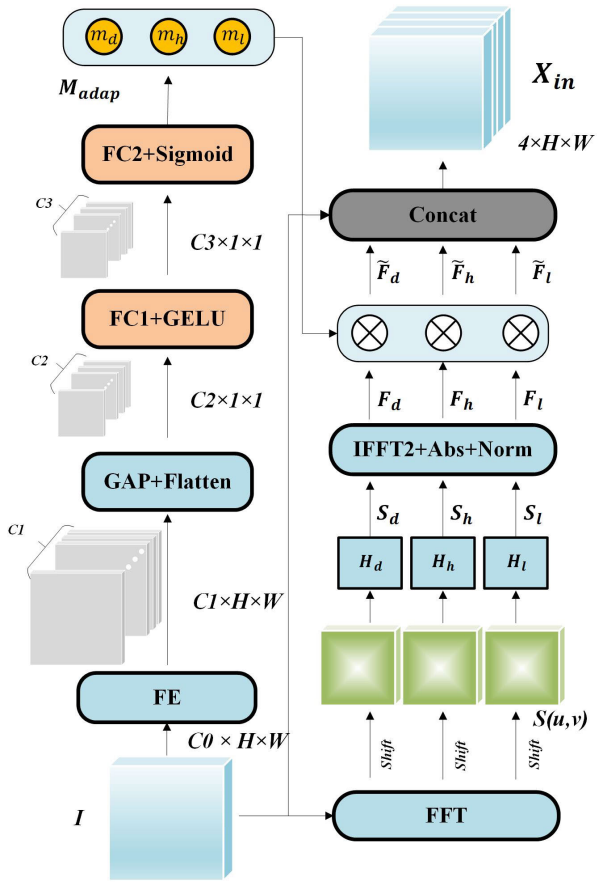


Fig. 5. Detailed architecture of the ASAM. The module extracts complementary frequency-domain priors by combining predefined filters with input-adaptive modulation. The input grayscale image I is transformed by FFT and decomposed into directional, high-frequency, and low-frequency components, yielding the auxiliary feature maps F_d, F_h, and F_l. The adaptive weight vector $M_{\text{adap}} = [m_d, m_h, m_l]$ recalibrates their contributions to generate the enhanced features $\widetilde{F}_d$, $\widetilde{F}_h$, and $\widetilde{F}_l$, which are concatenated with I to form a four-channel spectral–spatial input for the encoder.

H_l, are applied to the shifted spectrum to isolate different physical frequency priors.

$$S_k(u,v) = H_k(u,v) \odot S(u,v), \quad k \in \{d,h,l\} \tag{2}$$

where $\odot$ denotes elementwise multiplication. To accommodate the varying propagation directions of ISWs across different oceanic regions and eliminate orientation bias, H_d is designed as an omnidirectional star-like filter that uniformly partitions the angular Fourier domain into eight independent sectors. This allows the directional component to robustly capture anisotropic energy distributions without being restricted to a specific wave angle. Meanwhile, the high-frequency component captures fine edge textures, and the low-frequency component represents large-scale background variations. Each filtered spectrum is then shifted back and transformed into the spatial domain to obtain three auxiliary feature maps

$$F_k = \phi\left(\mathcal{F}^{-1}\left(\text{Shift}^{-1}(S_k)\right)\right), \quad k \in \{d,h,l\} \tag{3}$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the inverse FFT, $\text{Shift}^{-1}(\cdot)$ denotes the inverse frequency shift operation, and $\phi(\cdot)$ represents the spatial modulus calculation of the inverse-transformed complex spatial array followed by normalization.

Although these predefined filters provide useful physical frequency priors, purely static filtering operations are insufficient to adapt to dynamic oceanic imaging conditions, such as varying illumination, cloud interference, and complex sea-surface backgrounds. Therefore, ASAM further introduces a learnable modulation branch to dynamically adjust the contribution of the three filtered components according to the current input scene.

Specifically, the learnable modulation branch first applies a lightweight feature extraction block FE(·) to the original input image to obtain an intermediate feature representation

$$E = \text{FE}(I), \quad E \in \mathbb{R}^{C_1 \times H \times W}. \tag{4}$$

GAP and flattening are then performed on E to generate a compact global descriptor. This descriptor is passed through two fully connected layers with a GELU activation and a Sigmoid activation to produce an adaptive weight vector $M_{\text{adap}} \in \mathbb{R}^3$

$$M_{\text{adap}} = \sigma(\text{FC}_2(\text{GELU}(\text{FC}_1(\text{Flatten}(\text{GAP}(E)))))) \tag{5}$$

where GAP(·) denotes GAP, GELU(·) denotes the Gaussian error linear unit activation function, and σ represents the Sigmoid activation function that normalizes the modulation weights to the range of $(0, 1)$. The resulting adaptive weight vector can be written as

$$M_{\text{adap}} = [m_d, m_h, m_l] \tag{6}$$

where m_d, m_h, and m_l correspond to the directional, high-frequency, and low-frequency components, respectively.

After the predefined frequency-domain decomposition, the adaptive weight vector M_{adap} is used to recalibrate the contributions of the directional, high-frequency, and low-frequency feature maps

$$\widetilde{F}_k = m_k \cdot F_k, \quad k \in \{d,h,l\}. \tag{7}$$

Finally, the original spatial image and the three adaptively enhanced frequency-derived feature maps are concatenated along the channel dimension to form a four-channel heterogeneous input tensor

$$X_{\text{in}} = \text{Concat}\left(I, \widetilde{F}_d, \widetilde{F}_h, \widetilde{F}_l\right), \quad X_{\text{in}} \in \mathbb{R}^{4 \times H \times W}. \tag{8}$$

Through this design, ASAM combines predefined physical frequency priors with input-adaptive modulation. The static filters explicitly extract directional, high-frequency, and low-frequency energy components associated with ISW signatures, while the learnable modulation branch adjusts their relative importance under different oceanic background conditions. As a result, the generated four-channel input tensor provides complementary spectral–spatial representations for the subsequent backbone network.

2) Channel Attention Mechanism: While CNNs can extract rich spatial features, standard channel attention mechanisms, such as SE-Net, typically employ GAP to compress spatial information. From a signal processing perspective, GAP is merely equivalent to preserving the direct current (DC) component of the spectrum, while neglecting the high-frequency details that are crucial for delineating ISW textures. To overcome this information loss, we embed the frequency-domain

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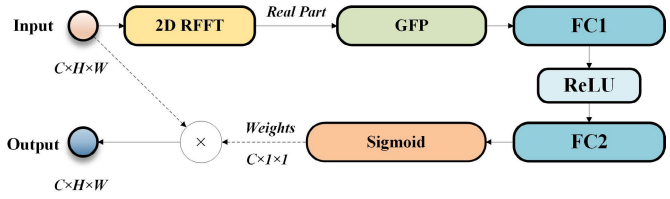


Fig. 6. FCA module schematic. Unlike GAP-based attention, which retains only DC components, FCA utilizes 2-D RFFT and GFP to aggregate global spectral information. It generates adaptive channel weights via nonlinear mapping to highlight periodic ISW textures, effectively mitigating high-frequency loss during downsampling.

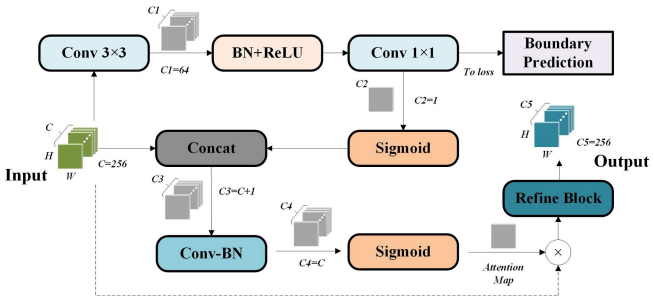


Fig. 7. Structural diagram of the BAM. The module processes the high-resolution feature maps from the BiFPN via a gated convolution mechanism. By generating an auxiliary boundary prediction map, it imposes explicit supervision on edge pixels.

channel attention (FCA) module within the network's feature extraction stage. As illustrated in Fig. 6, the FCA generalizes the traditional channel attention mechanism by introducing frequency-domain transformations, thereby achieving adaptive selection of multifrequency components along the channel dimension.

Specifically, given an input feature tensor $X \in \mathbb{R}^{C\times H\times W}$, the FCA first applies the 2-D FFT to project it into the frequency domain and extracts the real component of the spectrum to capture the significant physical energy representations of ISWs. Subsequently, the 2-D spectrum is compressed into a 1-D frequency descriptor via global frequency pooling (GFP). This step effectively preserves the periodic responses of ISW stripes at specific frequencies.

The compressed frequency vector undergoes nonlinear mapping via a two-layer fully connected network containing ReLU activation, ultimately generating a channel weight vector $W \in \mathbb{R}^{C\times 1\times 1}$ via a Sigmoid function. This weight is multiplied with the original feature X in a channelwise elementwise manner to complete feature recalibration. This process can be formally formulated as

$$X_{\text{out}} = X \odot \sigma\left(W_2\delta\left(W_1 \text{Real}\left(\text{FFT}\left(X\right)\right)\right)\right) \tag{9}$$

where Real$(\cdot)$ denotes the operation of extracting the real part, and δ and σ represent the ReLU and Sigmoid functions, respectively.

By introducing full spectrum perception, FCA can adaptively enhance feature channels containing periodic textures while suppressing random sea-surface noise channels, significantly improving the model's discriminative power regarding the physical characteristics of ISWs.

D. Cross-Scale Synergy and Boundary-Aware Refinement

To restore the refined geometric structures of ISWs while maintaining semantic consistency, a cross-scale collaborative architecture is designed within the decoding stage. Furthermore, the output quality is jointly constrained through a boundary-aware mechanism and a multitask loss function.

1) Global–Local Feature Evolution: We integrate a lightweight Transformer module at the top layer (C_5) of the encoder. In contrast to the local receptive fields of convolution operators, the Transformer leverages the self-attention mechanism to capture long-range dependencies across the global scope. This capability is pivotal for identifying ISW packets that span hundreds of kilometers. Subsequently, to facilitate the fusion of deep semantics and shallow details, the BiFPN is adopted. BiFPN transcends the limitations of unidirectional information flow inherent in traditional FPNs by establishing bidirectional pathways. By introducing learnable scalar weights, BiFPN dynamically adjusts the fusion contribution of features at different resolutions (C_1–C_5). This mechanism ensures that the faint textures of ISWs are not diluted within the deep network, while simultaneously allowing high-level semantics to precisely guide the localization of low-level edges.

2) BAM and Multitask Optimization: To address the fragmentation and boundary blurring of ISW signatures under complex sea-surface conditions, FSCNet adopts a dual-branch output architecture consisting of a primary semantic segmentation branch and an auxiliary boundary-aware branch. Due to the continuous bright-dark gradient transitions driven by sunglint modulation, ISW boundaries inherently lack physically sharp edges. Consequently, rather than relying on an independently annotated single-pixel physical edge, the auxiliary boundary target in BAM is defined as a morphological boundary band dynamically extracted from the binary ISW masks through the morphological gradient operation of dilation minus erosion. The BAM, illustrated in Fig. 7, is connected to the highest resolution fused feature map M_1 generated by the BiFPN. Through the gated convolution mechanism, BAM adaptively selects boundary-relevant responses from M_1 and produces a boundary probability map. This auxiliary branch provides explicit boundary-sensitive supervision and encourages the network to focus on ambiguous transition regions during optimization, thereby improving the topological continuity and boundary consistency of slender ISW structures.

To jointly optimize semantic segmentation and boundary detection, we define the overall multitask loss as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}}. \tag{10}$$

The main segmentation loss combines Dice loss, focal loss, derivative of union (DoU) loss, and Tversky loss

$$\mathcal{L}_{\text{main}} = 0.3\mathcal{L}_{\text{Dice}} + 0.5\mathcal{L}_{\text{Focal}} + 0.1\mathcal{L}_{\text{DoU}} + 0.1\mathcal{L}_{\text{Tversky}}. \tag{11}$$

Dice loss improves the global overlap between the predicted mask and the ground truth, while focal loss alleviates the severe foreground-background imbalance caused by sparse

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ISW pixels. Tversky loss introduces an asymmetric penalty for false positives and false negatives. In our implementation, the false-positive and false-negative weights are set to $\alpha_{tv} = 0.3$ and $\beta_{tv} = 0.7$, respectively, making the loss more sensitive to missed ISW pixels.

To further enhance boundary fitting, the DoU loss incorporates a boundary weight map generated from the ground truth mask

$$W = 1 + \alpha \cdot \text{Norm}\left(|\nabla^2 * Y_{\text{gt}}|\right)$$
$$\mathcal{L}_{\text{DoU}} = 1 - \frac{2\sum\left(Y_{\text{pred}} \odot Y_{\text{gt}} \odot W\right) + \epsilon}{\sum\left(Y_{\text{pred}} \odot W\right) + \sum\left(Y_{\text{gt}} \odot W\right) + \epsilon}. \tag{12}$$

Here, Y_{pred} and Y_{gt} denote the predicted probability map and the ground truth mask, respectively; $*$ denotes the convolution operation, $\odot$ denotes elementwise multiplication, and ϵ prevents zero division. By applying the Laplacian operator, the weight map W assigns higher weights to gradient inflection regions around ISW boundaries. However, instead of enforcing strict point-to-point edge matching, $\mathcal{L}_{\text{DoU}}$ computes a regionally weighted overlap between the predicted probability map and the ground truth mask. This formulation establishes a spatial tolerance region around diffuse ISW boundaries. Consequently, minor pixel-level shifts within the morphological transition band have limited influence on the weighted overlap, whereas macroscopic structural discontinuities along the wave skeleton substantially reduce the overlap score. This design reduces the sensitivity to local annotation uncertainty while preserving the topological continuity of elongated ISW signatures.

For the auxiliary boundary branch, the boundary loss $\mathcal{L}_{\text{aux}}$ is implemented using binary cross-entropy computed against the aforementioned morphologically derived boundary band, rather than against an independently annotated razor-thin single-pixel edge. This auxiliary supervision helps the model learn boundary-sensitive features and improves the structural consistency of the final segmentation results.

To systematically determine these hyperparameters, we formulated the loss-weight assignment as a black-box optimization problem and employed Bayesian optimization. Specifically, the weights of the loss components were selected using a randomly sampled subset containing 20% of the entire dataset for loss-weight determination and were rounded for stable implementation. The final configuration assigns higher weights to Focal loss and Dice loss to address foreground sparsity and maintain global structural consistency, while DoU loss, Tversky loss, and the auxiliary boundary loss provide additional boundary-aware and asymmetric supervision.

IV. Experiments Results and Analysis

A. Implementation Details and Evaluation Protocols

1) Experimental Environment and Parameter Settings: To ensure the fairness and reproducibility of the experimental results, all comparative experiments and ablation studies presented in this article were conducted within a unified hardware and software environment. The experimental platform was configured with an Intel[1] Xeon[1] Platinum 8352V CPU @ 2.10 GHz and an NVIDIA RTX A6000 GPU (48 GB VRAM). The software environment was built on the 64-bit Ubuntu 22.04.5 operating system, utilizing Python 3.9 and the PyTorch 2.1.0 deep learning framework, accelerated by CUDA 12.1 for computational tasks.

Regarding training strategies, input images are uniformly resized to 512×512 pixels, and the batch size is set to 8. We utilize the Adam optimizer with an initial learning rate of 0.0001. Furthermore, the Cosine Annealing strategy is employed to dynamically adjust the learning rate, thereby preventing the model from becoming trapped in local optima.

2) Quantitative Evaluation Metrics: Considering the characteristics of foreground sparsity and large aspect ratios exhibited by oceanic ISWs in imagery, relying solely on pixel accuracy fails to objectively reflect segmentation quality. Consequently, this study employs three core metrics widely accepted in the field of semantic segmentation for quantitative evaluation: Dice, mean IoU (mIoU), and mean accuracy (mAcc).

Dice: This metric represents the harmonic mean of precision and recall, effectively addressing the issue of positive–negative sample imbalance. Its calculation formula is as follows:

$$Dice = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN} \tag{13}$$

where TP, FP, and FN denote the number of true positives, false positives, and false negatives, respectively.

mIoU: This metric measures the degree of overlap between predicted masks and ground truth labels, serving as a critical indicator for evaluating the alignment of segmentation boundaries. Its calculation is defined as

$$\text{mIoU} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{TP_i}{TP_i + FP_i + FN_i} \tag{14}$$

where $k + 1$ denotes the total number of classes (including background), i is the class index, and the subscript i (e.g., TP_i) indicates the count for that specific class.

mAcc: This metric calculates the average classification accuracy across all classes and is utilized to evaluate the model's consistency in overall pixel classification

$$\text{mAcc} = \frac{1}{k+1}\sum_{i=0}^{k}\frac{TP_i}{TP_i + FN_i} \tag{15}$$

where the variables follow the same definitions as above.

B. Comparative Analysis With State-of-the-Art Methods

To comprehensively evaluate the robustness and advancement of the proposed model under complex sea conditions, this study establishes a benchmark suite covering representative segmentation paradigms from recent years. First, DeepLabV3 is selected as a classic spatial-context baseline to evaluate the effectiveness of atrous spatial pyramid pooling in capturing multiscale ISW patterns. Second, frequency-domain-aware and multiscale models, including FFTUNet, PAUNet [43], and MTU^2-Net, are introduced to examine the contribution of frequency transformation and multiscale representation

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TABLE I
PERFORMANCE AND EFFICIENCY COMPARISON WITH STATE-OF-THE-ART METHODS

Model	Dice↑	mIoU↑	mAcc↑	Params(M)↓	FLOPs(G)↓	Latency(ms)↓
DeepLabV3	0.5362	0.6809	0.8235	39.630	163.710	12.07
FFTUNet	0.5691	0.6970	0.8061	31.040	219.100	12.33
MTU^{2}Net	0.6085	0.7259	0.8159	175.190	177.450	33.75
PAUNet	0.6017	0.7128	0.7747	31.210	220.810	13.79
SwinUNet	0.5732	0.6992	0.7833	30.090	33.370	**8.95**
MVANet	0.5294	0.3958	0.7794	94.140	857.220	191.28
SLCNet	0.6392	0.7326	0.8069	187.147	350.001	34.71
UNetFormer	0.4681	0.6502	0.7325	11.700	**11.250**	56.66
RS3Mamba	0.3382	0.5947	0.7322	13.314	11.781	26.40
Ours	**0.6418**	**0.7352**	**0.8260**	**9.810**	117.20	29.98

to periodic texture discrimination. Third, Transformer-based, attention-enhanced, multitask, and remote-sensing-oriented architectures, including SwinUNet [46], MVANet [47], SLCNet [48], UNetFormer [49], and RS3Mamba [50], are further compared to assess the competitiveness of long-range dependency modeling, supervised correlation learning, global–local attention, and state-space modeling in complex ISW segmentation scenarios.

1) Quantitative Performance Assessment: As shown in Table I, FSCNet achieves the best overall segmentation performance among all compared methods, with a Dice score of 0.6418, an mIoU of 0.7352, and an mAcc of 0.8260. These results demonstrate that the proposed model provides the most favorable segmentation accuracy across both overlap-based and pixel-level metrics. Compared with conventional spatial-domain models and frequency-domain-aware methods, FSCNet shows clear advantages in Dice and mIoU, indicating its stronger capability to preserve the connectivity and edge integrity of slender ISW structures while suppressing interference from complex sea-surface backgrounds.

A detailed comparison further reveals the limitations of different architectural paradigms. DeepLabV3 obtains a relatively high mAcc of 0.8235, but its Dice score and mIoU are only 0.5362 and 0.6809, respectively. This high-accuracy but low-overlap behavior suggests that its predictions are strongly influenced by the dominant background class and that it struggles to capture sparse, narrow, and elongated ISW stripes. Frequency-aware or multiscale models, such as FFTUNet, PAUNet, and MTU^2-Net, improve the Dice and mIoU scores compared with DeepLabV3, demonstrating the usefulness of frequency information and multiscale representations for mitigating background interference. Nevertheless, their performance remains limited when dealing with weak, fragmented, and highly elongated ISW signatures, especially under complex sea states.

The newly introduced advanced benchmark models provide a more comprehensive comparison with recent remote-sensing segmentation paradigms. Among all baseline models, SLCNet achieves the strongest segmentation accuracy, with a Dice score of 0.6392 and an mIoU of 0.7326, which are close to those of FSCNet. This indicates that supervised long-range correlation modeling and adaptive receptive-field feature extraction are effective for remote sensing segmentation. However, SLCNet relies on a very heavy architecture, requiring 187.147 M parameters and 350.001 G FLOPs. In contrast, FSCNet slightly outperforms SLCNet in Dice, mIoU, and mAcc while using only 9.810 M parameters and 117.20 G FLOPs. This comparison demonstrates that FSCNet achieves a better tradeoff between segmentation accuracy and computational efficiency.

UNetFormer and RS3Mamba represent two recent efficient global-context modeling paradigms. UNetFormer adopts a lightweight CNN encoder and Transformer-based decoder to capture global–local contextual information, while RS3Mamba introduces visual state-space modeling for long-range dependency representation in remote sensing image segmentation. Although both models have very low computational complexity, with 11.250 and 11.781 G FLOPs, respectively, their segmentation accuracy on ISW extraction is clearly inferior to that of FSCNet. Specifically, UNetFormer obtains a Dice score of 0.4681 and an mIoU of 0.6502, while RS3Mamba obtains a Dice score of 0.3382 and an mIoU of 0.5947. These results suggest that efficient global dependency modeling alone is insufficient for ISW segmentation, where weak texture recovery, fine boundary localization, and continuity preservation of narrow wave fronts are equally critical.

Overall, FSCNet achieves the most favorable balance between segmentation accuracy and computational efficiency. It obtains the best Dice, mIoU, and mAcc scores while using only 9.810 M parameters, the smallest model size among all compared methods. Although SwinUNet, UNetFormer, and RS3Mamba show advantages in latency or FLOPs, their segmentation accuracy is clearly lower than that of FSCNet. In contrast, SLCNet achieves competitive accuracy but requires substantially more parameters and computational cost. With an inference latency of 29.98 ms, corresponding to approximately 33 FPS, FSCNet can meet the high-frequency observation requirements of geostationary satellite monitoring and provides a practical solution for real-time ISW segmentation.

2) Qualitative Visualization Analysis: To intuitively compare the segmentation behavior of different models under complex sea states, Fig. 8 presents qualitative results on six representative test samples. These samples cover diverse visual conditions, including clear wave signatures, weak contrast, complex sea-surface interference, and discontinuous wave patterns. The first four columns illustrate typical cases in

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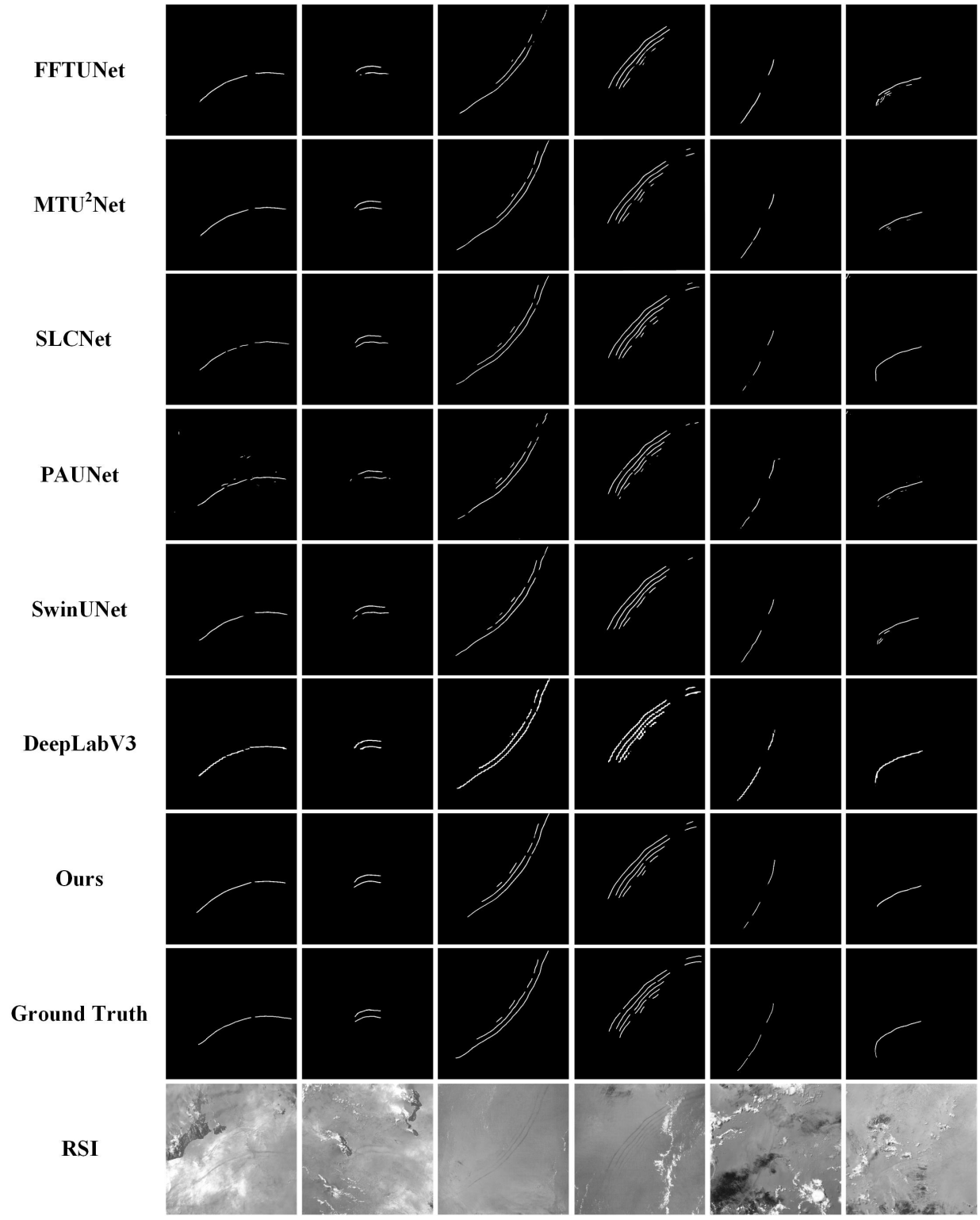


Fig. 8. Qualitative comparison of different segmentation models under representative sea states. Columns 1–4 show typical cases with relatively distinguishable ISW signatures, while Columns 5–6 show more challenging cases with weak, isolated, or discontinuous wave-like patterns under complex sea-surface backgrounds.

which the target structures are relatively distinguishable and are used to evaluate the ability of different models to preserve topological connectivity, boundary sharpness, and background noise suppression. The last two columns show more challenging cases with weak or discontinuous signatures, providing a visual analysis of the potential limitations of different methods.

As shown in the first four columns of Fig. 8, all models can roughly identify high-contrast ISW patterns when the wave signatures are distinct from the surrounding sea surface.

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TABLE II
PERFORMANCE AND COMPUTATIONAL EFFICIENCY COMPARISON OF CANDIDATE BACKBONE ARCHITECTURES

Model	Dice↑	mIoU↑	mAcc↑	Params(M)↓	FLOPs(G)↓	Latency(ms)↓
MobileNetV3	0.6204	0.7235	**0.8274**	29.02	121.48	58.97
ResNet	0.6166	0.7216	0.8224	128.83	164.22	35.95
SwinT	0.5636	0.6946	0.7901	30.47	**19.69**	24.23
ConvNeXt	0.4758	0.6543	0.7498	45.28	54.71	**15.89**
Ours	**0.6418**	**0.7352**	0.8260	**9.81**	117.20	29.98

TABLE III
PERFORMANCE COMPARISON OF THE FSCNET COMPONENT ABLATION VARIANTS

Different Variants	Model Components						
	V1	V2	V3	V4	V5	V6	V7
EfficientNet+BiFPN	✓	✓	✓	✓	✓	✓	✓
Static Frequency Channels		✓	✓	✓	✓	✓	✓
Light Transformer				✓	✓	✓	✓
FCA					✓	✓	✓
BAM+Multi-Loss						✓	✓
ASAM			✓				✓
Dice Score	0.5548	0.5804	0.5972	0.5856	0.5942	0.6027	0.6418

However, clear performance differences emerge under weak signal conditions and complex backgrounds. DeepLabV3 and SwinUNet tend to produce fragmented predictions and incomplete wave-train structures, indicating their limited ability to recover slender and spatially discontinuous ISW signatures. Frequency-aware and multiscale models, such as FFTUNet, PAUNet, and MTU^2-Net, improve the localization of ISW regions to some extent, but their predicted masks still suffer from coarse boundaries, local discontinuities, or residual background interference. SLCNet shows competitive visual performance among the baseline models and can preserve relatively complete wave structures in several cases. Nevertheless, its predictions still exhibit boundary ambiguity or local false responses under complex sea-surface textures.

In contrast, FSCNet provides more continuous and compact segmentation results in the typical success cases. Benefiting from the ASAM, the proposed model can exploit low-frequency structural constraints and directional frequency cues to reconstruct the continuous skeletons of ISW trains while suppressing nonperiodic sea-surface interference. Moreover, the boundary-aware refinement design further strengthens the response to high-frequency edge information, leading to sharper and smoother ISW boundaries that are more consistent with the ground truth.

The last two columns further illustrate challenging cases with weak or isolated wave signatures. In these samples, the ISW structures lack clear repetitive stripe patterns and exhibit relatively low contrast against complex sea-surface backgrounds. As a result, their frequency-domain responses are less regular and less distinguishable from background interference. Such isolated linear textures may be confused with cloud-shadow-induced patterns, noise, or spurious oceanic fronts, leading to fragmented predictions or missed detections. These results suggest that FSCNet performs more reliably on well-developed wave-packet or wave-train signatures, while weak, isolated, and nonperiodic ISW structures remain more challenging.

C. Ablation Studies on Architecture and Key Components

1) Ablation Study on Encoder Backbone Selection: To examine the influence of the encoder backbone, we compare EfficientNet with several representative alternatives, including MobileNetV3, ResNet, Swin Transformer, and ConvNeXt, under the same FSCNet framework and identical training settings. The quantitative results and efficiency metrics are reported in Table II.

As shown in Table II, the EfficientNet-based variant achieves the best overlap-based segmentation performance, with the highest Dice score of 0.6418 and mIoU of 0.7352, while requiring only 9.81 M parameters. Although MobileNetV3 obtains a slightly higher mAcc, its Dice and mIoU are lower, indicating weaker structural overlap for slender ISW targets. Swin Transformer and ConvNeXt show advantages in FLOPs or latency, but their segmentation accuracy decreases notably. ResNet also produces lower Dice and mIoU scores while introducing substantially more parameters and computational cost. These results indicate that EfficientNet provides the most favorable balance between segmentation accuracy and model efficiency, and is therefore adopted as the encoder backbone of FSCNet.

2) Ablation Studies on Key Components: To evaluate the contribution of each component in FSCNet, we conduct ablation experiments under a unified experimental setting. A network using EfficientNet and BiFPN is adopted as the baseline. In Table III, static frequency channels denote the four-channel input constructed by concatenating the original grayscale image with predefined frequency-derived auxiliary maps, whereas ASAM introduces input-adaptive spectral modulation on these frequency-derived features.

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As shown in Table III, the baseline model obtains a Dice score of 0.5548. After introducing static frequency channels, the Dice score increases to 0.5804, indicating that frequency-derived auxiliary information is beneficial for distinguishing weak ISW textures from complex sea-surface backgrounds. When ASAM is further added, the Dice score improves to 0.5972, showing that adaptive spectral modulation further enhances the effectiveness of frequency-domain representation.

The effects of spatial modeling and boundary refinement are evaluated in V4–V6. With static frequency channels as input, adding the lightweight Transformer improves the Dice score to 0.5856, suggesting that global spatial dependency modeling helps represent elongated ISW structures. The addition of FCA further increases the Dice score to 0.5942, indicating that frequency-aware channel recalibration contributes to more robust feature representation. After incorporating BAM and multitask loss, the Dice score reaches 0.6027, demonstrating the effectiveness of boundary-aware supervision in preserving slender ISW structures and improving edge localization.

The complete FSCNet achieves the highest Dice score of 0.6418. Compared with V6, V7 further incorporates ASAM and improves the Dice score from 0.6027 to 0.6418, indicating that adaptive frequency-domain modulation remains effective when combined with global dependency modeling, channel recalibration, and boundary-aware refinement. Overall, the ablation results show that the proposed components provide complementary benefits, leading to improved structural continuity and boundary localization for ISW segmentation.

D. Performance Analysis Across Morphological Categories

To further evaluate the robustness of FSCNet under different ISW morphologies, the test set is stratified according to geometric characteristics. Specifically, the samples are categorized into four primary morphological types: single wave, wave train, wave packet, and multipacket. Each primary type is further divided into three crest-connectivity states: continuous, segmented, and fragmented. Representative samples are shown in Fig. 9, where the number of samples in each subcategory is annotated in the upper right corner of each panel. Since no sample in the test set satisfies the definition of multipacket–continuous, this subcategory is not evaluated.

The sample distribution across morphological categories is imbalanced because visible-light ISW observations are strongly constrained by cloud cover, illumination, sunglint geometry, sea-surface background interference, and strict annotation quality control. Weak, fragmented, or partially obscured ISW signatures are therefore less frequently retained after screening. Subcategories with limited samples are included only when their ISW signatures are clearly distinguishable and manually verified. Therefore, the corresponding results are mainly intended to describe model behavior under different morphological conditions and should be interpreted with caution rather than as strong statistical comparisons.

Fig. 10 presents the categorywise quantitative results using raincloud plots, which combine probability density distributions, boxplots, and individual data points. Overall, FSCNet achieves relatively stable performance on well-developed

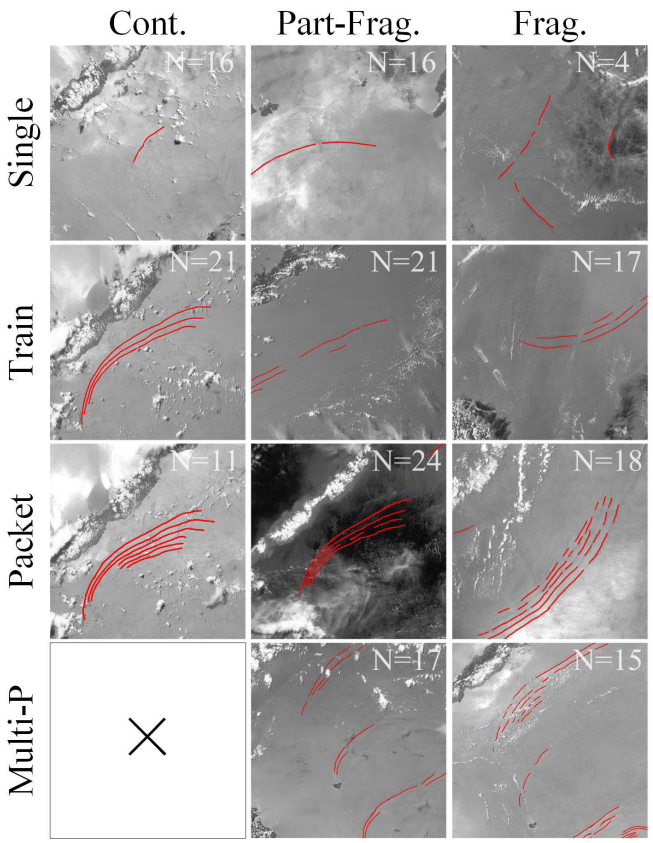


Fig. 9. Visualization of fine-grained ISW morphologies. The test set is stratified into four primary types based on physical structure, namely single wave, wave train, wave packet, and multipacket, and three subclasses based on crest connectivity, namely continuous, segmented, and fragmented. The sample count of each subcategory is annotated in the upper right corner of the corresponding panel. The "×" symbol indicates that no samples were found for the multipacket–continuous category.

wave-packet and multipacket samples. In particular, the multipacket–segmented category obtains the highest mean Dice score of 0.696. This result suggests that samples with multiple wave crests and clearer periodic structures provide more distinguishable frequency-domain cues, which can be effectively enhanced by the proposed ASAM.

An interesting observation is that, within the wave-train and wave-packet categories, fragmented samples can achieve performance comparable to or even higher than continuous samples. For example, the wave-packet–fragmented category obtains a mean Dice score of 0.639, exceeding the 0.580 of wave-packet–continuous. This may be because some fragmented samples still contain locally salient wave segments with strong gradients, which can be captured by the multiscale aggregation capability of BiFPN, even when long-range crestline continuity is partially disrupted.

In contrast, the single-wave–continuous category remains the most challenging case, with a lower mean Dice score of 0.569 and a more dispersed distribution. These samples usually appear as isolated, smooth, and weak linear signatures, lacking both repetitive spectral patterns and strong local gradients. As a result, they are more easily confused with oceanic fronts, ship wakes, or other linear sea-surface structures. This

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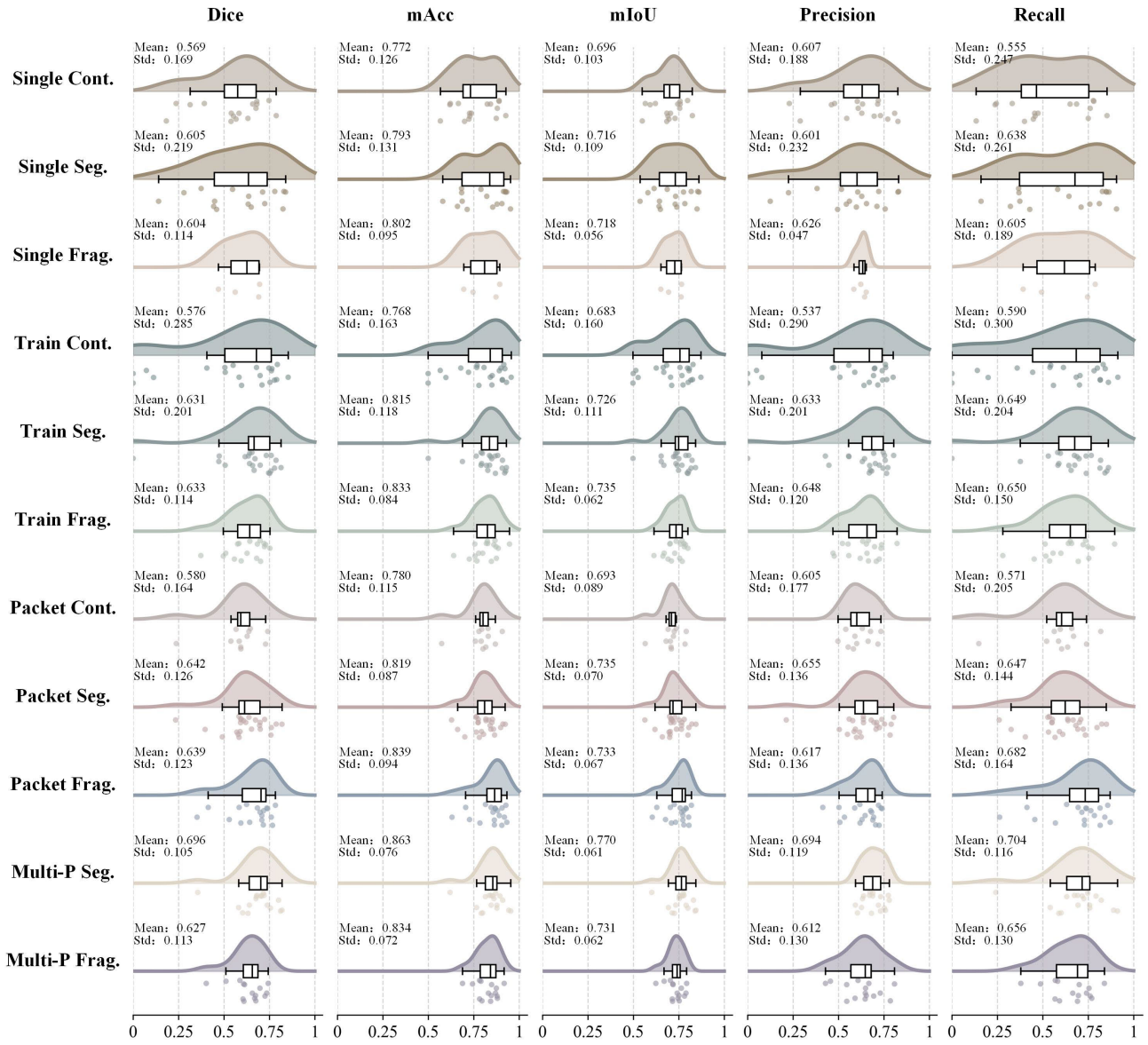


Fig. 10. Statistical distribution of performance metrics across distinct morphological categories. This visualization employs raincloud plots to display the specific distributions of five metrics (Dice, mAcc, mIoU, precision, recall) for each subclass. Each plot integrates a probability density estimation, a box plot representing quartiles, and jittered raw data points to reveal statistical dispersion. Text annotations indicate the mean and standard deviation (Std). Note: The multi–P Cont. category is omitted due to the absence of samples.

indicates that weak and isolated ISW signatures remain a major challenge for the current model and require further investigation.

In summary, although isolated continuous single-wave samples remain challenging, the cross-morphological evaluation demonstrates that FSCNet exhibits good robustness, particularly for ISW fields with complex physical structures and periodic dynamic features. Meanwhile, acknowledging the observational constraints and category-specific sample distributions helps ensure that the fine-grained evaluation remains physically grounded and statistically transparent.

E. Cross-Region and Cross-Sensor Generalization Analysis

To further evaluate the generalization ability of the proposed model beyond the Sulu sea and Sulawesi sea, we conduct an additional cross-region and cross-sensor experiment using a newly constructed Himawari-9 dataset over the Lombok Strait. This region is also a typical ISW-active area but differs from the original study regions in geographical structure, background texture, cloud distribution, and imaging conditions. Therefore, it provides a suitable test case for assessing whether the learned representation is transferable to unseen marine regions.

The additional dataset is derived from the AHI onboard the Himawari-9 satellite, with a spatial resolution of 500 m. A total of 52 image-mask pairs are manually constructed for this validation experiment. All models are trained only on the original FY-4 Sulu sea and Sulawesi sea dataset and directly evaluated on the Himawari-9 Lombok Strait dataset without fine-tuning or retraining, forming a strict cross-region and cross-sensor inference setting.

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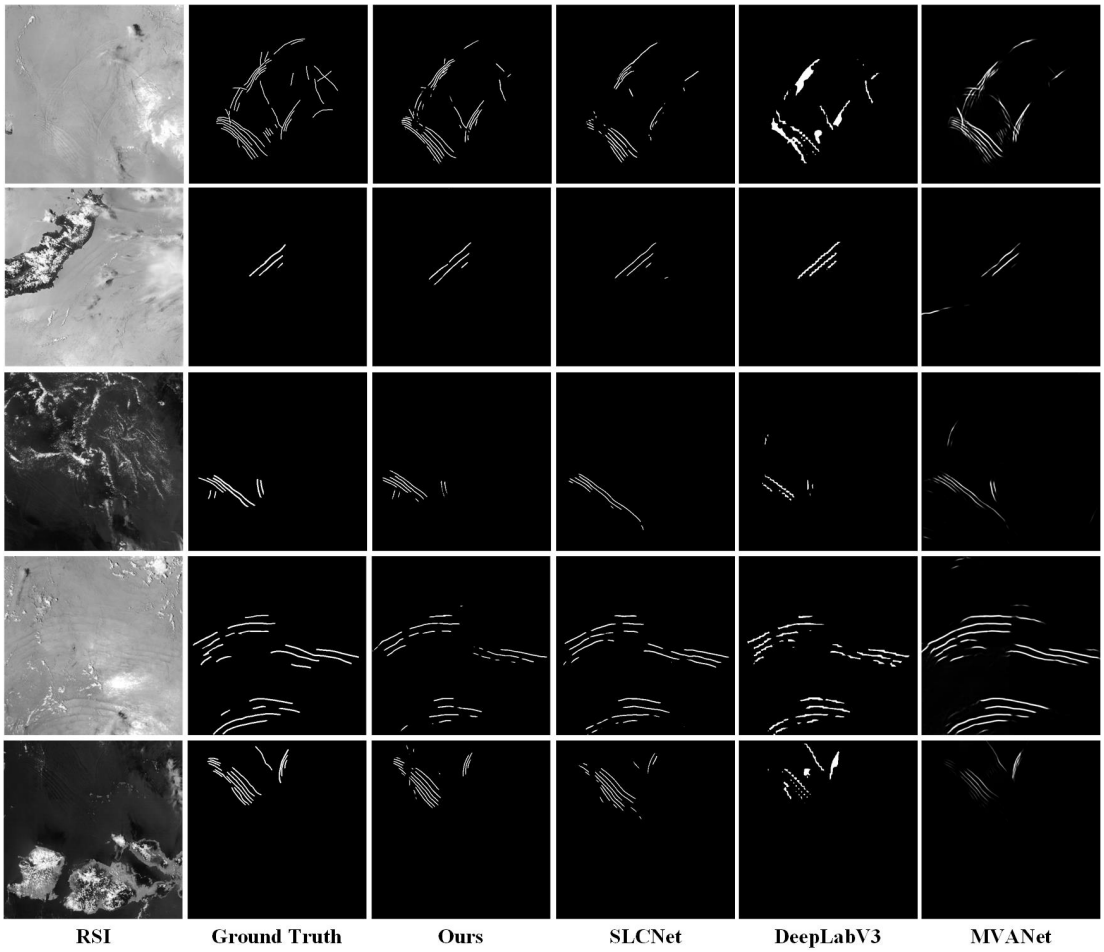


Fig. 11. Qualitative inference results of the top four models ranked by Dice score on the Himawari-9 Lombok Strait dataset. From left to right: RSI image, ground truth, ours, SLCNet, DeepLabV3, and MVANet.

TABLE IV
QUANTITATIVE INFERENCE RESULTS OF DIFFERENT MODELS ON THE HIMAWARI-9 DATASET

Model	Dice↑	mIoU↑	mAcc↑
DeepLabV3	0.4400	0.6460	0.7166
FFTUNet	0.4173	0.6435	0.6827
MTU^{2}Net	0.2260	0.5724	0.5927
MVANet	0.4951	0.6774	**0.7474**
PAUNet	0.2201	0.5666	0.5876
SwinUNet	0.2262	0.5724	0.5927
SLCNet	0.4733	0.6520	0.6893
UNetFormer	0.1453	0.5413	0.5694
RS3Mamba	0.1002	0.5253	0.5511
Ours	**0.5661**	**0.6986**	0.7384

As reported in Table IV, FSCNet achieves the best overall performance, with the highest Dice score of 0.5661 and mIoU of 0.6986. Although MVANet obtains a slightly higher mAcc, its Dice and mIoU are clearly lower, indicating less complete and less accurate ISW structure segmentation. These results demonstrate that FSCNet maintains stronger segmentation robustness under unseen-domain conditions.

The qualitative results in Fig. 11 further support this observation. The Lombok Strait samples contain complex cloud interference, island-induced textures, oceanic fronts, and weak wave signatures, which make the segmentation of slender ISW structures particularly challenging. Compared with DeepLabV3, MVANet, and SLCNet, FSCNet produces more continuous wave skeletons, better suppresses irrelevant background textures, and shows stronger consistency with the ground truth. This improvement can be attributed to the explicit frequency-domain enhancement of ASAM, which encourages the model to capture transferable wave-related periodic cues rather than region-specific background textures. Overall, the cross-domain experiment indicates that FSCNet has promising cross-region and cross-sensor generalization ability.

V. CONCLUSION AND FUTURE WORK

To address the challenges of weak signals, strong background interference, and variable cross-scale morphologies of oceanic ISWs in geostationary satellite visible-light imagery, this article proposes FSCNet, a frequency-spatial collaborative segmentation framework. By introducing ASAM, FSCNet explicitly enhances the frequency-domain characteristics of

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ISW signatures and suppresses stochastic sea-surface interference. In combination with BiFPN-based multiscale feature fusion and BAM-based boundary refinement, the proposed framework improves both the structural continuity and boundary localization of slender ISW patterns.

Experimental results on the standardized FY-4A/B dataset demonstrate that FSCNet achieves better segmentation performance than the compared mainstream models in terms of Dice and mIoU. The additional Himawari-9 Lombok Strait experiment further indicates its promising cross-region and cross-sensor generalization ability. The morphological category analysis shows that the proposed frequency-domain enhancement is particularly effective for well-developed ISW signatures with wave-packet structures or fragmented crest lines, where periodic stripe patterns and local gradients provide discriminative cues. However, isolated single-wave signatures with weak texture contrast and limited spectral regularity remain challenging, as they can be confused with oceanic fronts, ship wakes, or other linear sea-surface structures.

Future work will focus on extending FSCNet from single-frame segmentation toward dynamic and multisource ISW interpretation. First, the high-temporal-resolution observations of geostationary satellites will be exploited to develop spatiotemporal modeling methods for ISW tracking, propagation prediction, and source-region tracing. Such temporal information may help distinguish isolated ISW signatures from relatively static oceanic fronts or dissipating ship wakes. Second, multisource data fusion will be explored by incorporating complementary observations, such as SAR imagery, to provide additional physical constraints under weak periodicity or cloud-contaminated conditions. In addition, the constructed FY-4 dataset can be continuously expanded in spatial coverage, temporal span, and sensor diversity by incorporating observations from other ISW-active regions and geostationary satellites such as Himawari and GOES. These efforts are expected to support more robust cross-region and cross-sensor ISW segmentation, as well as downstream tasks such as detection, tracking, propagation prediction, and source-region tracing.

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